

Supplementary Material: Scenario Studies for Agentic UE-CoMIMO for 6G

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Abstract—The main article presents Agentic UE-CoMIMO as a vision for AI-native virtualized terminals and illustrates it with a system-level simulation of an AI-glass uplink live-streaming session across an outdoor-to-indoor (O2I) boundary. Because the magazine format cannot carry the full scenario, model, and parameter detail, this supplement documents that case study in full.

In *Scenario 1*, the glasses stream across that boundary and then dwell indoors for an extended session. A smartphone hub agent performs scene-aware band pre-switching, recruits a *shared* indoor customer-premises equipment (CPE) by choosing between full delegation (phone cellular radio off) and packet-split cooperation as the CPE background load varies, and runs a predictive energy-aware controller over the long dwell. Beyond the thermal/energy bitrate governor, the controller exploits the static environment and the known trajectory to predict scene complexity and stream content-adaptively: it holds the most quality-efficient target rung as an energy budget, lowers the encoded bitrate when the predicted scene is static, and adapts the capture frame rate to attack the camera power that the bitrate lever cannot reach, all at unchanged *perceived* quality. The control objective is quality of experience (QoE), including glass temperature, glass battery, and phone battery, rather than peak throughput, and we score it both per step and as a longevity-aware useful-quality-seconds figure that rewards sustaining target-quality video for longer. Across four trajectory families and twelve seeds under non-ideal prediction, the proposed policy attains both the best QoE and the most useful quality-seconds in every run.

We then study *Scenario 2*, a wearable blind-spot sensing episode in which a rear-approaching cyclist triggers a smartwatch body blockage and, later, a transient failure of the scooter CPE. Watch-only sensing fails the blockage and a static CPE commitment fails the failure, whereas the agentic hub recruits the CPE for the blockage and re-plans to phone-watch bistatic sensing on a micro-agent failure report, holding the rear-hazard warning through both events with about a fivefold higher joint QoE. For each scenario we give the narrative, the complete system model with all symbols defined, a fully sourced parameter table that separates standards-based from illustrative values, and a discussion of the simulation results. An interactive implementation accompanies this document, and every equation and number below matches that code.

I. PURPOSE AND SCOPE

This supplement complements the main article in two ways. First, it records the engineering detail (geometry, channel

This document is supplementary material for the article “Agentic UE-CoMIMO for 6G: From Virtual Antenna Augmentation to AI-Native Virtualized Terminals.” It provides the scenario narrative, system model, parameter settings, and full simulation discussion that the magazine format cannot accommodate. Each scenario is self-contained, and additional scenarios will be appended as separate sections. The reference simulation code and interactive browser implementations of both scenarios are available at <https://chaokaiwen.github.io/agentic-ue-comimo/>.

and device models, control logic, and parameters) behind the agentic-control evaluations, at a level of rigor appropriate for reproduction but inappropriate for a magazine. Second, it serves as a growing library of *scenarios*, where each section defines one self-contained case study that exercises a particular aspect of Agentic UE-CoMIMO. The condensed illustration in the main article and the full scenario here both target the agentic control plane (UE-CoMIMO 2.0), which sits on top of the physical-layer virtual-antenna mechanisms (UE-CoMIMO 1.0): the main article reports the headline result, and this supplement gives the reproducible detail.

Throughout, we separate two classes of parameters. **Standards-based** values are taken from 3GPP, IEEE 802.11, IEC, or ITU-T references and are marked accordingly. **Illustrative** values are engineering choices selected to expose a qualitative trade-off; their absolute magnitudes are plausible but are not measured device figures, and the conclusions are stated to depend only on their ordering, not on their exact values.

II. SCENARIO 1: AI-GLASS UPLINK LIVE STREAMING ACROSS AN O2I BOUNDARY

A. Narrative and Motivation

A creator wearing AI glasses live-streams first-person video while walking from an outdoor street into an indoor venue such as a mall, studio, or exhibition hall, then settles at a fixed spot and keeps streaming for an extended session. The glasses generate an *uplink* video flow, so this is not a downlink consumption problem. The glasses have limited RF and thermal headroom, so they offload video over 5 GHz Wi-Fi to a smartphone and never transmit at 700 MHz, because a low-band antenna is hard to integrate into an eyeglass form factor. The smartphone is the hub agent. It carries the cellular uplink, runs lightweight scene understanding, and decides how the cluster cooperates.

Two difficulties drive the scenario. The first is the O2I transition. Outdoors, the 3.5 GHz uplink offers ample capacity. As the user approaches and crosses the entrance, the 3.5 GHz link suffers heavy building penetration loss and a transient doorway blockage, and can collapse within a second or two. A purely reactive controller only switches bands *after* the link has degraded, paying a reconnection gap on an already-failing link and stalling the live stream. Indoors, a 700 MHz fallback cannot sustain the target bitrate through a high-loss building, so the hub agent engages a nearby indoor CPE. Worse, parts of the venue are outright RF dead-zones (a shielded lift lobby or a metal-clad hall) where the macro

link to the gNB vanishes entirely, so a reactive band-switch can only stall, and the stream survives only by delegating to the in-venue CPE. The second difficulty is the long indoor dwell itself. The CPE is *shared* infrastructure whose spare uplink capacity fluctuates with other users, the smartphone battery drains fastest when its cellular transmitter is active, and the glasses heat toward their skin-contact limit if the encoder keeps running at the top bitrate. The control objective is the creator's QoE, namely visual quality, stall avoidance, latency, glass battery and temperature, and phone battery, rather than peak throughput. This scenario therefore exercises five agentic behaviors at once: intent-aware band selection, scene-aware proactive (make-before-break) switching, task-driven recruitment of an assisting device, resource-rational delegation to shared infrastructure (turning the phone's cellular radio off when the CPE alone suffices), and thermal foresight (relaxing the bitrate before the temperature limit is reached, rather than after). Over the long dwell it adds three predictive energy-aware behaviors that turn foresight about the static scene into battery life: it rations energy by holding the most quality-efficient bitrate as a budget rather than chasing the top rung, it lowers the encoded bitrate when it predicts the scene will be static (which looks identical at far fewer bits), and it lowers the capture frame rate on a predicted-static dwell to cut the fixed camera power that no bitrate choice can touch.

B. Geometry and Mobility

Time is discretized with step $\Delta t = 0.1$ s over an episode of $T_{\text{ep}} = 900$ s, long enough to expose both the glass thermal dynamics (time constant ≈ 150 s, Section II-E) and the divergence in battery drain across policies. A single gNB sits at horizontal coordinate $x_g = -100$ m at height $h = 25$ m. The user walks along a straight line at speed $v = 1$ m/s and stops at an indoor spot $x_{\text{stop}} = 10$ m, so the user position is $x(t) = \min(x_0 + vt, x_{\text{stop}})$ with $x_0 = -30$ m. The entrance is at $x = 0$ (crossed at $t = 30$ s), the fully indoor region begins at $x = 5$ m (reached at $t = 35$ s), and the user dwells at x_{stop} from $t = 40$ s onward. The 3D user-to-gNB distance is $d(t) = \sqrt{(x(t) - x_g)^2 + h^2}$. An indoor CPE near a window is placed at $x_c = 15$ m and becomes discoverable once the user reaches the entrance; its gNB distance $d_c = \sqrt{(x_c - x_g)^2 + h^2}$ is constant. The indoor crossing fraction and indoor depth are

$$p_f(t) = \text{clip}(x(t)/5, 0, 1), \quad (1)$$

$$\delta(t) = \max(0, x(t) - 5), \quad (2)$$

where $\text{clip}(z, 0, 1) = \min(\max(z, 0), 1)$. The AI glasses always reach the smartphone over a short 5 GHz Wi-Fi link that is provisioned not to be the bottleneck. The cluster uses one of five transmission modes, summarized in Table I.

C. Channel Model

All distances are in meters and all power quantities in dB or dBm. The path loss follows the 3GPP TR 38.901 urban-macro (Ura) NLOS model

$$\text{PL}(d, f) = 13.54 + 39.08 \log_{10}(\max(d, 1)) + 20 \log_{10}(f), \quad (3)$$

TABLE I
TRANSMISSION MODES IN SCENARIO 1.

Mode	Path	Description
M0	glass $\rightarrow$ gNB	AI-glass direct 3.5 GHz uplink (fallback only, high glass power and heat)
M1	glass $\rightarrow$ phone $\rightarrow$ gNB	Wi-Fi offload, smartphone relays on 3.5 GHz (outdoor workhorse)
M2	glass $\rightarrow$ phone $\rightarrow$ gNB	Wi-Fi offload, smartphone relays on 700 MHz (indoor robustness)
M3	glass $\rightarrow$ phone + CPE $\rightarrow$ gNB	Non-coherent packet-split cooperation on 700 MHz across smartphone and CPE (Option C)
M4	glass $\rightarrow$ phone $\rightarrow$ CPE $\rightarrow$ gNB	Full delegation: phone relays over Wi-Fi to the CPE and turns its cellular radio off (phone battery saver)

with carrier frequency f in GHz. The receiver noise power on a band of width B (in Hz) is

$$N(B, I) = -174 + 10 \log_{10}(B) + \text{NF} + I, \quad (4)$$

where $\text{NF} = 7$ dB is the noise figure and I is an interference margin, set to $I_{700} = 6$ dB on the busier 700 MHz band and $I = 0$ otherwise. The three bandwidths are $B_w = 80$ MHz (Wi-Fi) and the per-user cellular shares $B_{3.5} = 50$ MHz and $B_{0.7} = 10$ MHz, each half of the underlying 100 and 20 MHz carriers (Table II).

The O2I penetration loss ramps with $p_f(t)$ using a 3GPP TR 38.901 *high-loss* building model that is consistent across bands:

$$L_{\text{pen}}^{3.5}(t) = L_w^{3.5} p_f(t) + L_{\text{door}} [4 p_f(t)(1 - p_f(t))], \quad (5)$$

$$L_{\text{pen}}^{0.7}(t) = L_w^{0.7} p_f(t), \quad (6)$$

with steady wall losses $L_w^{3.5} = 28$ dB and $L_w^{0.7} = 18$ dB. The bracketed factor $4p_f(1 - p_f)$ equals 1 at mid-crossing ($p_f = 0.5$) and 0 at both ends, so $L_{\text{door}} = 8$ dB is the peak transient doorway blockage (body and door-frame diffraction), applied to 3.5 GHz only. Indoor clutter adds $L_{\text{cl}}(t) = \min(0.2\delta(t), 4)$ dB. Each mobile link further carries a random term $X_b(t)$ (for band $b \in \{3.5, 0.7\}$) that is the sum of two components: spatially correlated log-normal shadowing with standard deviations $\sigma_{3.5} = 4$ dB and $\sigma_{0.7} = 3$ dB and a 10 m decorrelation distance, plus a temporal first-order autoregressive fading component with standard deviation $\sigma_t = 1.5$ dB and a 2 s correlation time, which models people moving around the user; the channel therefore keeps fluctuating after the user stops walking. Body loss is $L_b^g = 2$ dB for the glasses and $L_b^p = 3$ dB for the smartphone.

Finally, a shielded interior pocket (for example a lift lobby or a metal-clad hall) over the position interval $x \in [x_{d0}, x_{d1}] = [8, 14]$ m adds a *dead-zone* loss

$$L_{\text{dz}}(t) = L_d \mathbf{1}\{x(t) \in [x_{d0}, x_{d1}]\}, \quad L_d = 60 \text{ dB}, \quad (7)$$

to the *outdoor macro* uplink only (both the 3.5 GHz and the 700 MHz links to the gNB). The in-venue CPE, reached over the local Wi-Fi hop and backhauled on its own fixed link, is unaffected. The dead-zone thus blocks the cellular-only policies while leaving CPE delegation intact, isolating a

coverage-hole regime from the thermal and energy regime that governs the rest of the episode.

With transmit power $P_g = 20$ dBm and antenna gain $G_g = -2$ dBi for the glasses, and $P_p = 23$ dBm, $G_p = 0$ dBi for the smartphone, the three mobile-link SINRs (in dB) are

$$\gamma_g(t) = P_g + G_g - \text{PL}(d, 3.5) - X_{3.5} - L_{\text{pen}}^{3.5} - L_{\text{cl}} - L_b^g - L_{\text{dz}} - N(B_{3.5}, 0), \quad (8)$$

$$\gamma_p^{3.5}(t) = P_p + G_p - \text{PL}(d, 3.5) - X_{3.5} - L_{\text{pen}}^{3.5} - L_{\text{cl}} - L_b^p - L_{\text{dz}} - N(B_{3.5}, 0), \quad (9)$$

$$\gamma_p^{0.7}(t) = P_p + G_p - \text{PL}(d, 0.7) - X_{0.7} - L_{\text{pen}}^{0.7} - L_{\text{cl}} - L_b^p - L_{\text{dz}} - N(B_{0.7}, I_{700}). \quad (10)$$

The CPE link is treated as a *static* link with its own fixed wall loss $L_c = 12$ dB and no shadowing, clutter, or body loss, since the CPE is fixed near a window:

$$\gamma_c = P_c + G_c - \text{PL}(d_c, 0.7) - L_c - N(B_{0.7}, I_{700}), \quad (11)$$

with $P_c = 23$ dBm and $G_c = 10$ dBi (a fixed directional panel antenna). The efficiency-discounted achievable rate of a link with SINR γ (dB), efficiency η , and bandwidth B is

$$R(\gamma; \eta, B) = \eta B \log_2(1 + 10^{\gamma/10}), \quad (12)$$

with implementation efficiencies $\eta_c = 0.55$ (cellular) and $\eta_w = 0.50$ (Wi-Fi). The individual link rates are $R_g = R(\gamma_g; \eta_c, B_{3.5})$, $R_p^{3.5} = R(\gamma_p^{3.5}; \eta_c, B_{3.5})$, and $R_p^{0.7} = R(\gamma_p^{0.7}; \eta_c, B_{0.7})$. The Wi-Fi rate uses $\gamma_w = 25$ dB outdoors and 22 dB indoors, $R_w = R(\gamma_w; \eta_w, B_w)$.

The CPE is *shared* infrastructure. Its total uplink grant is $C_{\text{tot}} = 40$ Mbps, and a two-state Markov background load $\beta(t)$, representing other users of the venue, occupies 5 Mbps (mean dwell 30 s) or 25 Mbps (mean dwell 15 s). The capacity allocatable to our cluster is therefore

$$R_c(t) = \min(R(\gamma_c; \eta_c, B_{0.7}), C_{\text{tot}} - \beta(t)) \in \{35, 15\} \text{ Mbps}, \quad (13)$$

since the physical CPE link rate (≈ 38 Mbps at $\gamma_c \approx 21$ dB) exceeds the grant, so the CPE backhaul can carry the target rung on its own. This fluctuation is what forces the agent to keep re-balancing between the delegation and cooperation modes indoors; inside the dead-zone of (7), where the phone's own 700 MHz leg is blocked, the cooperative split (M3) loses its second link and collapses onto pure CPE delegation (M4), so the strong CPE backhaul becomes the only path that keeps the stream alive.

D. Cooperative and Delegated Uplink Modes

Packet-split cooperation (M3) follows the spec Option C non-coherent hybrid: a fraction $\alpha = 0.8$ of the traffic is split across the smartphone and CPE 700 MHz links while the remainder is duplicated on the stronger link, minus a coordination overhead $O = 2$ Mbps:

$$R_3 = \max\left(0, \alpha(R_p^{0.7} + R_c) + (1 - \alpha)\max(R_p^{0.7}, R_c) - O\right). \quad (14)$$

Because the split operates at the packet level (multi-path scheduling, in the spirit of multi-path transport, not PHY combining), no additional spectral-efficiency penalty is applied; the entire coordination cost is the explicit overhead O .

Full delegation (M4) routes everything through the CPE: the glasses offload to the phone over Wi-Fi as usual, the phone forwards over a second Wi-Fi hop to the CPE, and the CPE alone transmits at 700 MHz, so the phone's cellular radio is switched off entirely. The two Wi-Fi hops are assumed to operate on separate channels and do not contend for airtime, so

$$R_4 = R_c. \quad (15)$$

Because the glasses offload to the smartphone, every relayed mode is also bounded by the Wi-Fi rate. The delivered rate of each mode is therefore

$$R_{M0} = R_g, \quad R_{Mk} = \min(R_w, R_k), \quad k \in \{1, 2, 3, 4\}, \quad (16)$$

where $R_1 = R_p^{3.5}$ and $R_2 = R_p^{0.7}$. Modes M0, M1, and M2 are always available; M3 and M4 require the CPE, i.e. $x(t) \geq 0$.

E. Traffic, Energy, and Thermal Models

The video source uses adaptive bitrate (ABR) streaming over the ladder $\mathcal{B} = \{3, 8, 15, 25\}$ Mbps, corresponding respectively to a low-quality fallback, HD live video, high-quality HD/2K live video, and premium 2K/near-4K live video. The target rung is set to $r_{\text{tgt}} = 15$ Mbps, which represents a high-quality live-streaming operating point that avoids the disproportionate thermal cost of the top rung. A rate margin $\Delta r = 3$ Mbps is imposed to reduce buffer depletion. Given the delivered rate $R_{\text{del}}(t)$ of the active mode, the selected bitrate $r_v(t)$ is the highest $b \in \mathcal{B}$ satisfying $R_{\text{del}}(t) \geq b + \Delta r$. The governor in Section II-F may cap the available ladder at r_{tgt} . In addition, the selected bitrate is dropped by one rung when the buffer falls below 1 s and is forced to the lowest rung when $T > T_{\text{max}}$.

The live buffer has a capacity $B_{\text{cap}} = 2.5$ s, reflecting that a live sender cannot pre-deliver an arbitrarily long future segment, and is initialized at 2.0 s. Its evolution is modeled as

$$B_{\text{uf}}(t + \Delta t) = \min\left(B_{\text{cap}}, \max\left(0, B_{\text{uf}}(t) + \Delta t(R_{\text{del}}(t)/r_v(t) - 1)\right)\right), \quad (17)$$

where $R_{\text{del}}(t)/r_v(t) > 1$ increases the buffer and $R_{\text{del}}(t)/r_v(t) < 1$ drains it. A stall occurs when $B_{\text{uf}}(t) = 0$.

1) *Scene complexity and content-adaptive coding*: A static, low-motion first-person view reaches a given *perceived* quality at a far lower encoded bitrate, and a far lower frame rate, than a fast-moving one. We model the instantaneous scene complexity $C(t) \in [0, 1]$ as a smoothed sum of ego-motion, the O2I doorway transition, and nearby crowd motion,

$$C_{\text{raw}}(t) = \text{clip}(C_0 + w_m m(t) + w_d \mathbf{1}\{\text{doorway}\} + w_c \rho(t), 0, 1), \quad (18)$$

where $m(t) = \mathbf{1}\{|x_{\text{tgt}} - x(t)| > 0\}$ flags walking toward the intended stop x_{tgt} , $\rho(t) \in \{0, 0.5, 1\}$ is the crowd level (idle,

passers-by, busy CPE spell), and the value is low-pass filtered as $C(t) = a_c C(t - \Delta t) + (1 - a_c) C_{\text{raw}}(t)$ with $a_c = e^{-\Delta t / \tau_c}$. With $C_0 = 0.10$, $w_m = 0.60$, $w_d = 0.50$, $w_c = 0.40$, and $\tau_c = 1$ s, a quiet indoor dwell sits at $C \approx 0.1$, while walking, crossing the doorway, or a passing crowd drive $C \rightarrow 1$.

The bitrate and frame rate needed to reach the target perceived quality scale with complexity,

$$r_{\text{need}}(C) = r_{\text{tgt}}(\rho_r + (1 - \rho_r)C), \quad (19)$$

$$\phi_{\text{need}}(C) = \phi_0 + (1 - \phi_0)C, \quad (20)$$

with floors $\rho_r = 0.40$ and $\phi_0 = 0.60$ (a 15 fps minimum capture), so a static scene needs only ≈ 6 Mbps at half the frame rate whereas full motion needs the full 15 Mbps at 30 fps. The *perceptual* rate utility used in the QoE score (Section II-F) replaces the raw bitrate ratio with the weaker of the bitrate and frame-rate adequacies,

$$u(r_v, \phi, C) = \min\left(\min\left(1, \frac{r_v}{r_{\text{need}}(C)}\right), \min\left(1, \frac{\phi}{\phi_{\text{need}}(C)}\right)\right) + w_o \frac{\max(0, r_v - r_{\text{need}}(C))}{r_{\text{max}} - r_{\text{need}}(C)}, \quad (21)$$

where $\phi \in [\phi_0, 1]$ is the capture frame-rate fraction (relative to 30 fps). Under-provisioning either the bitrate or the frame rate for the current motion lowers u (visible as blocking or judder), whereas a static scene streamed at 8 Mbps and 20 fps and a moving scene at 15 Mbps and 30 fps both reach $u = 1$. For all baseline policies C is irrelevant ($\phi \equiv 1$ and the utility reduces to the saturating bitrate ratio at r_{tgt}); only the proposed policy exploits C through Mechanisms 2 and 3 below.

2) *AI-glass power and temperature*: With the normalized bitrate $\hat{r} = r_v / r_{\text{max}}$ and $r_{\text{max}} = 25$ Mbps (the top ladder rung), the glass power draw combines the camera, a superlinear encoder term, and the active radio,

$$P(t) = \phi(t)(P_{\text{cam}} + P_{e0}) + P_{e1} \hat{r}^{1.5} + P_{\text{link}}, \quad (22)$$

with $P_{\text{cam}} = 0.7$ W, encoder base $P_{e0} = 0.1$ W and slope $P_{e1} = 1.6$ W. The exponent 1.5 reflects that encoder power scales with pixel rate, so the top rung is disproportionately expensive. The frame-rate fraction $\phi(t)$ scales the camera and per-frame encoder base (sensor and ISP capture cost), whereas the bitrate-driven encoder term and the radio are data-volume and link costs that ϕ does not change; baselines run $\phi \equiv 1$. The radio term is $P_{\text{link}} = P_{\text{cell}} = 2.0$ W in the direct mode M0, and $P_{\text{link}} = P_{w0} + P_{w1} \hat{r}$ with $P_{w0} = 0.3$ W and $P_{w1} = 0.3$ W in every offloaded mode. A crucial consequence for longevity is that the *fixed* overhead $P_{\text{fix}} = P_{\text{cam}} + P_{e0} + P_{w0} = 1.1$ W is independent of bitrate; at the target rung it is 54% of the offloaded draw and at 8 Mbps it rises to 74%. The bitrate lever therefore saturates, and only reducing the capture frame rate (Mechanism 3) can attack the camera share of this fixed floor. A first-order thermal model gives the glass temperature,

$$T(t + \Delta t) = T_{\text{amb}} + a(T(t) - T_{\text{amb}}) + bP(t), \quad (23)$$

with ambient $T_{\text{amb}} = 28$ °C, initial temperature $T_{\text{amb}} + 6 = 34$ °C, and coefficients $a = 0.99933$ and $b = 0.004$ per 0.1 s step, which correspond to a thermal time constant $\tau = -\Delta t / \ln a \approx 150$ s and a steady-state thermal resistance

$b/(1 - a) = 6$ °C/W. The 150 s time constant gives the glasses a realistic multi-minute warm-up rather than an instantaneous jump. The safe and emergency skin-contact limits are $T_{\text{safe}} = 42$ °C and $T_{\text{max}} = 48$ °C. Two steady states anchor the thermal story: streaming the top rung over Wi-Fi draws 3.0 W, whose steady-state temperature 46.0 °C *violates* the safe limit, whereas the target rung draws 2.02 W and settles at 40.1 °C, safely below it. The thermal feasibility boundary therefore lies between the top two ladder rungs, which is exactly the headroom the governor exploits.

3) *Smartphone power*: The smartphone power is composed of additive states: a baseline $P_{\text{id}} = 0.3$ W, Wi-Fi reception from the glasses $P_{\text{rx}} = 0.4$ W (active in every relayed mode), the cellular transmitter $P_{\text{t35}} = 2.0$ W at 3.5 GHz or $P_{\text{t07}} = 1.2$ W at 700 MHz, and Wi-Fi forwarding to the CPE $P_{\text{tx,w}} = 0.6$ W. The per-mode totals are

$$P_{\text{ph}} = \begin{cases} 0.3 \text{ W} & \text{M0 (phone not in the path),} \\ 2.7 \text{ W} & \text{M1,} \\ 1.9 \text{ W} & \text{M2,} \\ 2.5 \text{ W} & \text{M3,} \\ 1.3 \text{ W} & \text{M4 (cellular radio off).} \end{cases} \quad (24)$$

M4 is the key entry: by delegating the whole uplink to the CPE, the phone drops to roughly half the draw of any mode that keeps its cellular transmitter alive. Phone energy is tracked as a KPI and enters the QoE score below.

4) *Battery and longevity*: Both devices carry finite batteries that drain at their instantaneous power. The AI glasses hold $E_g = 0.8$ Wh, the order of magnitude of a current streaming-class smart-glass (a 210 mAh cell sustains about 45 min of continuous capture in measured reviews), and the smartphone holds $E_{\text{ph}} = 15$ Wh (a typical 4000 mAh handset). Each device's remaining charge evolves as $E(t + \Delta t) = E(t) - P(t) \Delta t / 3600$ (Wh). If the glass battery reaches zero the stream ends: that creator is removed and its QoE average is *frozen* at the moment of death, so the headline metric measures QoE while actually streaming, and battery longevity is reported separately as remaining charge and uptime. Because the glass battery is small relative to a 0.8 to 4.4 W draw, longevity becomes a first-class objective over a long dwell. A policy that streams a higher bitrate buys instantaneous quality at the cost of a shorter session, so an agent optimizing the whole session must ration energy rather than chase rate.

F. Scene-Aware Prediction and Agentic Control

The hub agent predicts indoor entry from on-device scene understanding. With the logistic function $\sigma(z) = 1/(1 + e^{-z})$, the indoor probability is a sigmoid of position triggered 6 m before the entrance, passed through a 1 s inference delay τ_{inf} and additive Gaussian noise $\nu(t) \sim \mathcal{N}(0, 0.05^2)$:

$$p_{\text{in}}(t) = \text{clip}\left(\sigma(k(x(t - \tau_{\text{inf}}) - x_{\text{trig}})) + \nu(t), 0, 1\right), \quad (25)$$

with sharpness $k = 0.8$ and trigger position $x_{\text{trig}} = -6$ m.

1) *Thermal- and battery-aware bitrate governor*: The proposed policy alone runs a governor that caps the ABR ladder using three signals; the other policies leave the ladder uncapped (only the hard protection $T > T_{\max}$, forcing the lowest rung, applies to all). (i) *Predictive intent trigger*. The agent reasons about the *future* thermal trajectory: from (23) the steady-state temperature at glass power P is $T_{ss}(P) = T_{\text{amb}} + \frac{b}{1-a}P$, and the governor engages when the user has been fully indoors for at least 10 s, the buffer is at least 80% full, the session intent is a long stream, and T_{ss} at the top rung exceeds $T_{\text{safe}} - 0.5$ °C; it releases when the scene module signals departure ($p_{\text{in}} < 0.4$). A myopic QoE maximizer would not make this trade, because the thermal penalty below is zero until T_{safe} is crossed, by which point the heat is already stored. (ii) *Reactive thermal cap*. Independently of location, so that it also protects an outdoor long dwell, the ladder is capped at r_{tgt} once $T > T_{\text{safe}} - 1$ °C and at 8 Mbps once $T > T_{\text{safe}}$. (iii) *Energy-budget tier (Mechanism 1)*. The fixed-overhead structure of (22) makes the target rung the most quality-efficient operating point: because $P_{\text{fix}} = 1.1$ W flows regardless of bitrate, the quality delivered per joule is maximized at $r_{\text{tgt}} = 15$ Mbps (dropping below it loses rate utility faster than it saves power, while exceeding it burns power for the negligible w_o bonus). The proposed policy therefore caps the ladder at r_{tgt} at all times and *rides it to natural battery death*, rather than the reactive step-downs (to 8 and 3 Mbps near empty) of a naive battery saver: limping at 3 Mbps near the end buys a little runtime but at near-useless quality, lowering the integrated useful quality of Section II-F4. At r_{tgt} the glasses draw ≈ 2.0 W, giving a ≈ 24 min ceiling that no bitrate policy can exceed without sacrificing the target quality. Together these keep the proposed glasses within the thermal limit and maximize useful streaming life, whereas the ablation C, lacking the governor, does neither.

2) *Predictive complexity-aware streaming (Mechanisms 2 and 3)*: Because the environment is static, the channel as a function of position is a deterministic map known to the agent, and the user's near-future positions follow from the intended trajectory. The proposed policy exploits this by predicting the worst-case scene complexity over a horizon $H = 4$ s. Walking the predicted path $\{x(t+k\Delta t)\}_{k=1}^{H/\Delta t}$ and reading the deterministic ego-motion and doorway terms (with the current crowd level held as a conservative estimate), it forms $\hat{C}(t) = \max_k C(x(t+k\Delta t))$. The agent is *not* clairvoyant: $\hat{C}$ is corrupted by a Gaussian estimation error, $\tilde{C} = \text{clip}(\hat{C} + \epsilon, 0, 1)$ with $\epsilon \sim \mathcal{N}(0, \sigma_p^2)$ and $\sigma_p = 0.15$, and the result is low-pass filtered (a 3 s exponential moving average) so the encoder holds a rung rather than tracking the per-step noise.

Mechanism 2 (perceptual-rate bitrate). Rather than always streaming r_{tgt} , the policy encodes only what the predicted scene needs: the ladder is additionally capped at the lowest rung $b \in \mathcal{B}$ with $b \geq r_{\text{need}}(\tilde{C})$ from (19). A predicted static dwell ($\tilde{C} \approx 0.1$, $r_{\text{need}} \approx 7$ Mbps) thus streams at 8 Mbps, perceptually identical to 15 Mbps by (21) but at lower encoder power, while predicted motion keeps the full 15 Mbps.

Mechanism 3 (adaptive frame rate). The capture frame-rate fraction is matched to the predicted motion, $\phi(t) =$

$\phi_{\text{need}}(\text{clip}(\tilde{C} + \beta_\phi, 0, 1))$ from (20), with a small conservative bias $\beta_\phi = 0.08$ so that a misprediction errs toward smoother video rather than judder. On a static dwell this drops the camera to ≈ 20 fps, attacking the fixed P_{fix} overhead that the bitrate lever cannot reach; on motion it returns to 30 fps. The β_ϕ margin is deliberately decoupled from the bitrate choice, so the encoder still captures the full $8 \rightarrow 15$ Mbps energy saving while the frame rate stays on the safe side of judder.

3) *Mode selection*: The controller is a priority policy that operationalizes the intent-aware utility of the main article. The rate requirement follows the current quality demand: $r_{\text{req}} = b_{\text{top}} + \Delta r$, where b_{top} is the highest currently permitted rung (25 Mbps normally, 15 Mbps under the governor), giving 28 or 18 Mbps respectively:

- 1) indoors with the CPE available, choose among $\{M4, M2, M3\}$, ordered by *increasing phone power* (24), the first mode whose delivered rate meets r_{req} ; switching to a lower-power mode additionally requires a 5 Mbps hysteresis margin to prevent load-driven ping-pong; if no mode qualifies, pick the highest-rate one (best effort);
- 2) else if $p_{\text{in}} \geq 0.65$ or the 3.5 GHz SINR is no longer usable ($\gamma_p^{3.5} \leq 0$), *pre-switch* to 700 MHz (M2) before the link collapses;
- 3) else if outdoors with a usable 3.5 GHz link, use the smartphone 3.5 GHz relay (M1).

Rule 1 is the resource-rational core: when the shared CPE has headroom, M4 wins and the phone's cellular radio sleeps; when the CPE gets busy, M4 can no longer meet r_{req} and the controller escalates to M3, whose packet split adds the phone's own 700 MHz capacity on top of the loaded CPE. All band switches use *make-before-break*: during the 1.5 s reconnection gap (plus a one-time 1.0 s setup gap the first time the CPE is engaged), the old link keeps serving, so a proactive switch costs little while a reactive switch pays the gap on an already-collapsed link. A minimum mode dwell of 2 s further damps oscillation.

4) *QoE score and useful-quality longevity*: The per-step QoE score balances perceptual quality against stalls, switching delay, glass radio energy, phone energy, and thermal stress:

$$q(t) = w_r u(r_v, \phi, C) \mathbf{1}\{B_{\text{uf}}(t) > 0\} - w_s \mathbf{1}\{\text{stall}\} - w_d \frac{\tau_m}{0.2} - w_e \frac{P_{\text{rad}}}{P_{\text{cell}}} - w_{\text{ph}} \frac{P_{\text{ph}}}{P_{\text{ph}}^{\max}} - w_{\text{th}} \frac{\max(0, T - T_{\text{safe}})}{T_{\max} - T_{\text{safe}}} - w_{\text{sw}} \mathbf{1}\{\text{switch}\}, \quad (26)$$

where $\mathbf{1}\{\cdot\}$ is the indicator function, and the $\mathbf{1}\{B_{\text{uf}}(t) > 0\}$ multiplier on the quality term sets the reward to zero whenever the playback buffer is empty ($B_{\text{uf}} = 0$), so that a stalled frame receives no video-quality reward. The quality term is the *perceptual* utility $u(r_v, \phi, C)$ of (21), which saturates at 1 once the bitrate *and* frame rate meet the scene's complexity-dependent need $r_{\text{need}}(C)$, $\phi_{\text{need}}(C)$, and adds only a small bonus, weighted by $w_o = 0.1$, for exceeding it up to $r_{\max} = 25$ Mbps. For a baseline policy ($\phi \equiv 1$, and C unused so $r_{\text{need}} = r_{\text{tgt}}$) this reduces to the familiar saturating ratio at the

15 Mbps (2K-live) target. Crucially, scoring *perceived* rather than encoded quality is what lets the proposed policy bank energy on a static scene at 8 Mbps and 20 fps with no QoE loss, while a thermally throttled policy that drops to 8 Mbps during *motion* is correctly penalized. $P_{\text{rad}} = P_{\text{link}}$ is the glass *radio* power from (22) (the encoder's cost is already charged through the thermal term, so it is not double-counted here), $P_{\text{ph}}^{\text{max}} = 2.7$ W is the largest per-mode phone power (M1), and τ_m is the per-mode switching delay (in seconds), with $\tau_m \in \{0.02, 0.03, 0.03, 0.05, 0.05\}$ for M0 through M4. The weights are $w_r = 1$, $w_o = 0.1$, $w_s = 4$, $w_d = 1.5$, $w_e = 0.8$, $w_{\text{ph}} = 0.6$, $w_{\text{th}} = 3$, $w_{\text{sw}} = 0.5$. This ordering ($w_s > w_{\text{th}} > w_d > w_r > w_e > w_{\text{ph}} > w_{\text{sw}}$) encodes that a stall is the dominant QoE killer, consistent with ITU-T P.1203 and measurement studies of streaming QoE, and that exceeding the on-face skin-safety limit is the next most serious outcome, a hard constraint for a wearable rather than a soft preference, which is why w_{th} ranks just below w_s . The score is accumulated only while the device is alive; after a battery death it is frozen, as noted in Section II-E4.

Because average QoE alone is blind to how *long* a creator keeps delivering, and can even reward dying early to stop accumulating a slightly negative score, we adopt a longevity-aware figure of merit, the *useful quality-seconds*, that integrates the *saturating* perceived quality over the watchable session,

$$Q_{\text{use}} = \int_0^{T_{\text{ep}}} \tilde{u}(r_v, \phi, C) \mathbf{1}\{\text{alive}\} \mathbf{1}\{B_{\text{uf}} > 0\} dt, \quad (27)$$

where $\tilde{u} = \min(\min(1, r_v/r_{\text{need}}(C)), \min(1, \phi/\phi_{\text{need}}(C))) \in [0, 1]$ is the min-of-adequacies term of (21) *without* the over-target bonus, so the quality credit caps at 1 per second and Q_{use} can never exceed the alive, unstalled time. This single number captures both of the stated goals: it rewards higher perceived quality and a longer stall-free life, it saturates the quality credit at the target (so it does not over-reward chasing 4K), and it is never increased by dying sooner. It directly encodes the value the proposed policy adds, namely sustaining target-quality video for longer, and is used as the headline verdict alongside average QoE in Section II-I.

G. Baseline Policies

The proposed policy is compared against two fixed-or-reactive baselines and one ablation, all evaluated on the *same* precomputed channel and CPE-load traces for a paired comparison. (Earlier revisions also included a fixed-3.5 GHz relay and a fixed-700 MHz relay; these were transitional variants of the cases below, and since a real handset already band-switches they are dropped.)

- **A:** AI-glass direct 3.5 GHz (M0 fixed), the worst thermal case;
- **B:** reactive 700 MHz switch on a SINR threshold, where the smartphone starts on the 3.5 GHz relay (M1) and moves to 700 MHz (M2) only after an exponentially smoothed $\gamma_p^{3.5}$ falls below -5 dB (smoothing time constant 3.5 s);
- **C (ablation):** the identical scene-aware mode orchestration as the proposed policy, including M4 delegation, but

without the predictive energy-aware extensions, that is, no thermal/energy governor and no complexity-aware bitrate or frame-rate adaptation, so its ABR keeps chasing the top rung at full frame rate indoors. C isolates the value of the predictive control plane from that of the mode orchestration it shares with the proposed policy.

H. Parameter Settings

Table II lists all parameters with their provenance. The path-loss model, noise floor, O2I penetration, UE power class, Wi-Fi and NR bandwidths, and skin-contact thermal limits are standards-based, whereas device power, thermal dynamics, cooperation overhead, control thresholds, and QoE weights are illustrative.

I. Results and Discussion

We report results at two levels. Table III and Figs. 1–5 document the full proposed controller (all predictive mechanisms of Section II-F2 enabled, non-ideal prediction $\sigma_p = 0.15$) in detail on the canonical 900 s walk-in-and-dwell episode at seed 7, across the O2I transition and the indoor dwell. Table IV then evaluates the same controller against all baselines over four trajectory families and 12 seeds each, where the small glass battery is exercised to depletion, scored by perceived quality (21) and useful quality-seconds (27).

Table III reports the key metrics over the 900 s episode (random seed 7). Figure 1 shows the delivered bitrate and smoothed QoE across the O2I transition and the long indoor dwell, and Figs. 2 to 5 show the mode timeline, glass temperature, battery drain, and the aggregate comparison. The proposed policy spends 26 s in M1, 9 s in M2, and 865 s in M4; because the dwell lies inside the dead-zone, where the phone's 700 MHz leg is blocked and the packet split (M3) collapses onto pure delegation, it delegates fully to the CPE throughout the indoor dwell.

The results support ten claims, all consistent with the physical setup and robust to the illustrative parameters.

- 1) **The capability constraint is enforced.** The AI glasses never transmit at 700 MHz, and only the smartphone and CPE do. Mode M0 exists solely as a thermal- and battery-worst fallback.
- 2) **Direct cellular on the glasses overheats and drains fastest.** Policy A reaches 45.1 °C, above the 42 °C safe limit, and is down to 8.8% battery by $t = 900$ s, the lowest of all, because the glasses both encode and transmit at high power. Offloading the uplink off the glasses is therefore a thermal and battery necessity, not just a capacity choice.
- 3) **Direct 3.5 GHz on the glasses stalls through the boundary.** Policy A loses the link on entry and stalls for about 864 s of the 900 s episode, collapsing to the lowest bitrate and the worst QoE (-5.84); only offloading the uplink and adapting the band avoids it.
- 4) **A reactive radio cannot escape the macro dead-zone.** Two limits compound here. First, even where its macro link survives, 700 MHz alone cannot meet the

TABLE II
SCENARIO 1 PARAMETERS AND THEIR PROVENANCE. ILLUSTRATIVE VALUES ARE ENGINEERING CHOICES; CONCLUSIONS DEPEND ON THEIR ORDERING, NOT THEIR EXACT MAGNITUDES.

Group / parameter	Value	Basis
<i>Geometry and mobility</i>		
gNB position x_g / height h	−100 m / 25 m	3GPP TR 38.901 UMa geometry (10 m to 5 km valid range)
User speed v / start x_0 / stop x_{stop}	1 m/s / −30 m / 10 m	Pedestrian; entrance at $t = 30$ s, dwell from $t = 40$ s
Episode length T_{ep} / step Δt	900 s / 0.1 s	Long dwell exposes thermal ($\tau \approx 150$ s) and battery drain
CPE position x_c / discovery	15 m / from entrance	Illustrative (near-window placement)
<i>Radio and channel</i>		
Bandwidths $B_w/B_{3.5}/B_{0.7}$	80 / 50 / 10 MHz	IEEE 802.11ax; 3GPP NR n78 / n28 (cellular = per-user share, half the carrier)
Noise figure NF	7 dB	Typical UE receiver
Efficiency η_c/η_w	0.55 / 0.50	Illustrative (Shannon-to-implementation gap); M3 split is packet-level, no extra PHY penalty
Wall loss $L_w^{3.5}/L_w^{0.7}$	28 / 18 dB	3GPP TR 38.901 §7.4.3 high-loss building (consistent)
Doorway peak L_{door} (3.5 GHz)	8 dB transient	Illustrative (body and door-frame diffraction)
Indoor clutter	0.2 dB/m, cap 4 dB	Illustrative (cf. 3GPP O2I indoor loss)
Shadowing $\sigma_{3.5}/\sigma_{0.7}$	4 / 3 dB, corr. 10 m	3GPP TR 38.901 UMa NLOS (conservative)
Temporal fading σ_t / corr. time	1.5 dB / 2 s	Illustrative (people moving near the user)
Interference margin I_{700}	6 dB	Illustrative (busier low band)
Body loss L_b^p/L_b^g	3 / 2 dB	Illustrative
CPE wall loss L_c	12 dB, static link	Illustrative (fixed near-window CPE)
RF dead-zone $[x_{d0}, x_{d1}]$ / loss L_d	[8, 14] m / 60 dB	Illustrative (shielded interior pocket; macro only, CPE exempt)
Wi-Fi SINR γ_w (out / in)	25 / 22 dB	Illustrative (short link, not the bottleneck)
<i>Devices</i>		
Tx power $P_p/P_c/P_g$	23 / 20 dBm	3GPP TS 38.101 Power Class 3 (glass reduced)
Antenna gain $G_g/G_p/G_c$	−2 / 0 / 10 dBi	Illustrative (form-factor dependent; CPE directional panel)
Glass power $P_{cam}/P_{e0}/P_{e1}/P_{w0}/P_{w1}/P_{cell}$	0.7/0.1/1.6/0.3/0.3/2.0 W	Illustrative; superlinear encoder (exponent 1.5), Eq. (22)
Phone power $P_{id}/P_{tx}/P_{t35}/P_{t07}/P_{tx,w}$	0.3/0.4/2.0/1.2/0.6 W	Illustrative; per-mode totals 0.3 to 2.7 W, Eq. (24)
Thermal a/b	0.99933 / 0.004	Illustrative ($\tau \approx 150$ s, 6 °C/W)
Thermal limits $T_{amb}/T_{safe}/T_{max}$	28 / 42 / 48 °C	IEC 62368-1 / EN 563 skin-contact limits
Battery E_g/E_{ph}	0.8 / 15 Wh	Streaming-class glasses (~210 mAh); 4000 mAh phone
CPE grant C_{tot} / bg. load / mean dwell	40 / {5, 25} Mbps / 30 / 15 s	Illustrative (shared venue CPE, two-state Markov load)
Coop overhead O / setup / α	2 Mbps / 1 s / 0.8	Illustrative (Option C packet split)
<i>Traffic and control</i>		
Bitrate ladder $\mathcal{B}$ / target r_{tgt}	{3, 8, 15, 25} / 15 Mbps	Illustrative (720p to 4K-live ladder)
Buffer init / cap / low	2.0 / 2.5 / 1.0 s	Low-latency live streaming
p_{in} threshold	0.65	Illustrative
Reactive threshold / smoothing	−5 dB / 3.5 s	Illustrative (baseline B)
Min dwell / switch gap / power hysteresis	2.0 s / 1.5 s / 5 Mbps	Illustrative (gap $\gg$ real NR sub-100 ms, conservative)
Scene trigger x_{trig} / k / τ_{inf}	−6 m / 0.8 / 1 s	Illustrative (on-device inference latency)
Governor intent trigger: dwell / buffer / margin / exit	10 s / 80% / 0.5 °C / $p_{in} < 0.4$	Illustrative (predictive thermal governor)
Governor reactive thermal cap	$r_{tgt}@41^\circ\text{C}$, 8 Mbps@42°C	Illustrative (location-agnostic thermal tier)
Energy-budget tier (Mech. 3)	cap at r_{tgt} , ride to death	Illustrative (max quality-per-joule under fixed P_{fix})
Scene complexity $C_0/w_m/w_d/w_c/\tau_c$	0.10/0.60/0.50/0.40/1 s	Illustrative (ego-motion, doorway, crowd)
Perceptual floors ρ_r/ϕ_0	0.40/0.60	Illustrative (content-adaptive coding, 15 fps min)
Prediction horizon / noise / smoothing / fps bias	$H=4$ s / $\sigma_p=0.15/3$ s EMA / $\beta_\phi=0.08$	Illustrative (non-ideal trajectory/complexity prediction)
QoE weights $w_{r,s,d,e,ph,th,sw}/w_o$	1, 4, 1.5, 0.8, 0.6, 3, 0.5 / 0.1	Illustrative; ordering per ITU-T P.1203, thermal a hard wearable constraint

target: with 18 dB penetration and only a 10 MHz per-user share, the reactive policy (B) is held to the lowest rungs (average delivered ≈ 3.8 Mbps), never reaching the 15 Mbps target. Second, inside the shielded pocket the macro link is blocked outright (7), and because B's band-switch only ever reaches the smartphone 700 MHz and never the CPE, it stalls for 860 s of the dwell (QoE −4.57), almost as badly as the direct-3.5 GHz baseline A. A reactive band-switch is helpless in a coverage hole; escaping it needs a proactive delegation to alternative infrastructure (the CPE), which the scene-

aware policies C and P make and which carries them through the hole with zero stalls.

- 5) **Proactive beats reactive because of make-before-break.** The proposed policy leaves 3.5 GHz 4.5 s before entry (the scene prediction and an early shadowing dip both push it off the band), whereas the reactive policy (B) switches 5.5 s after entry, paying the reconnection gap on a collapsing link. The proposed transition minimum QoE (−1.23) is a brief dip, whereas the direct-3.5 GHz policy A falls to −5 and stays there.
- 6) **The CPE makes the target bitrate available, and**

TABLE III

SCENARIO 1 RESULTS FOR THE *full* PROPOSED CONTROLLER (ALL PREDICTIVE MECHANISMS OF SECTION II-F2 ENABLED; QoE IS THE PERCEPTUAL UTILITY (21)), SEED 7, 900 S EPISODE, ENTRANCE AT $t = 30$ S, INDOOR DWELL FROM $t = 40$ S. “AVG BITRATE” IS THE AVERAGE *delivered video* BITRATE (THE PROPOSED POLICY ENCODES AT 8 MBPS ON THE STATIC DWELL, SO ITS AVERAGE SITS BELOW THE 15 MBPS TARGET WHILE ITS PERCEIVED QUALITY STAYS AT THE TARGET). “GLASS” AND “PHONE” ARE THE BATTERY CHARGE REMAINING AT $t = 900$ S. THE BEST PRACTICAL POLICY IS IN BOLD. “SWITCH LEAD” IS THE TIME BY WHICH THE CELLULAR BAND CHANGES BEFORE ENTRY, SO POSITIVE IS PROACTIVE AND NEGATIVE IS REACTIVE; A NEVER LEAVES 3.5 GHz (M0), WHILE B SWITCHES REACTIVELY AFTER THE LINK HAS ALREADY DEGRADED. THE INDOOR DWELL FALLS INSIDE THE RF DEAD-ZONE OF (7), WHICH BLOCKS THE MACRO 3.5/700 MHz BANDS, SO THE CELLULAR-ONLY BASELINES A AND B STALL THERE, WHEREAS THE SCENE-AWARE POLICIES C AND P DELEGATE TO THE IN-VENUE CPE AND STREAM THROUGH.

Policy	Avg bitrate (Mbps)	Stall (s)	Avg QoE	Min QoE (transition)	Max temp (°C)	Glass (%)	Phone (%)	Switch lead (s)
A: glass 3.5 direct	3.75	864	−5.84	−5.02	45.1	8.8	99.5	n/a
B: reactive 700 switch	3.78	860	−4.57	−4.78	36.3	60.4	96.8	−5.5
C: scene-aware w/o governor	19.65	0	−0.08	−1.23	44.2	21.2	97.8	+4.5
Proposed: scene-aware UE-CoMIMO	8.78	0	+0.10	−1.23	36.1	58.2	97.8	+4.5

the encoder uses only what the scene needs. The proposed policy spends 865 s on the CPE (here almost entirely on M4, since the dead-zone blocks the phone’s 700 MHz and collapses the M3 split onto pure delegation), whose boosted backhaul can sustain the 15 Mbps target throughout the dwell, headroom the form-factor-limited cluster cannot reach on the phone’s 700 MHz link alone. The encoder then streams at the target during motion and the busy spells and content-adaptively at 8 Mbps on the static dwell (Mechanism 2), with zero stalls and only brief dips at mode-switch instants, so the average delivered bitrate (8.8 Mbps) sits below the target while the perceived quality stays at it.

- 7) **Delegation puts the phone’s cellular radio to sleep.** Whenever the shared CPE has headroom the controller picks M4 (1.3 W phone draw) over M3 (2.5 W) and M2 (1.9 W). Inside the dead-zone, where the phone’s own 700 MHz is unusable anyway, full delegation is also the only mode that keeps the stream alive. The proposed policy spends its whole indoor dwell in M4 and uses 1214 J of phone energy, against 1740 J for the reactive baseline B, which keeps its 700 MHz transmitter running uselessly into the dead-zone. The ablation C, likewise forced onto the CPE, draws the same 1214 J, so among the streaming policies delegation roughly halves the phone radio cost relative to keeping a cellular transmitter alive. The instantaneous M4 saving is 48% relative to M3.
- 8) **The agent adapts to shared infrastructure.** Where the phone’s 700 MHz leg is available (outside the dead-zone, as in the indoor-pacing family of Table IV), the two-state CPE load drives an M3/M4 alternation: under light load (35 Mbps allocatable) the cluster delegates fully (M4), and when other users occupy the CPE (15 Mbps) the controller escalates to the packet split (M3), adding the phone’s own 700 MHz capacity through the busy bursts. In the canonical dwell this lever is unavailable, because the dead-zone blocks the 700 MHz leg, so the cluster relies on the CPE backhaul alone, which is precisely why a CPE strong enough to carry the target on its own (the 10 dBi directional panel) matters.
- 9) **The governor wins on QoE, safety, and longevity at once.** Without it (ablation C), chasing the top rung on

the strong CPE backhaul pushes the glasses past the safe limit during the dwell (peaking at 44.2 °C, Fig. 3) and drains the glass battery to 21.2% by $t = 900$ s (Fig. 4). With the full predictive controller, the proposed policy holds 36.1 °C (safe), retains 58.2% battery (about 2.8 times C’s remaining life), *and* achieves the best average QoE (+0.10 versus C’s −0.08). Beyond the safe limit the thermal penalty outweighs C’s bitrate advantage, and the complexity-aware coding further cools and powers down the glasses at no perceptual cost, so the controller yields a net QoE win while respecting the wearable’s hard thermal and battery constraints. A myopic QoE maximizer cannot make this trade, because the thermal penalty activates only after the limit is crossed and the battery cost is paid only later.

- 10) **Quality trades against session longevity on a fixed energy budget.** Glass battery at $t = 900$ s orders the streaming-quality policies by their encoder load: the ablation C (≈ 20 Mbps) ends at 21.2%, while the proposed policy (content-adaptive, 8.8 Mbps average) ends at 58.2% (Fig. 4). The reactive baseline B coasts to $\approx 60\%$ only because it is stalling at near-zero delivered rate in the dead-zone, not because it is streaming efficiently. Streaming more bits than the scene needs buys nothing in perceived quality but shortens the session, which is precisely why an agent optimizing the whole session rations energy through the energy budget and the complexity-aware bitrate and frame-rate, spending it only where the picture benefits.

These claims are qualitative. The reported numbers correspond to the fixed seed 7; the conclusions follow from designed regime boundaries (the penetration budget, the two CPE load states, the thermal steady states straddling T_{safe} , and the small glass battery relative to the streaming power) rather than from a particular noise realization. The accompanying interactive implementation reproduces the same ordering for any walk the user draws with the mouse.

J. Multi-Trajectory Evaluation of the Predictive Extensions

The canonical episode above fixes one walk-in trajectory. To stress the predictive extensions we evaluate the full controller over four trajectory families, each repeated for 12 seeds and

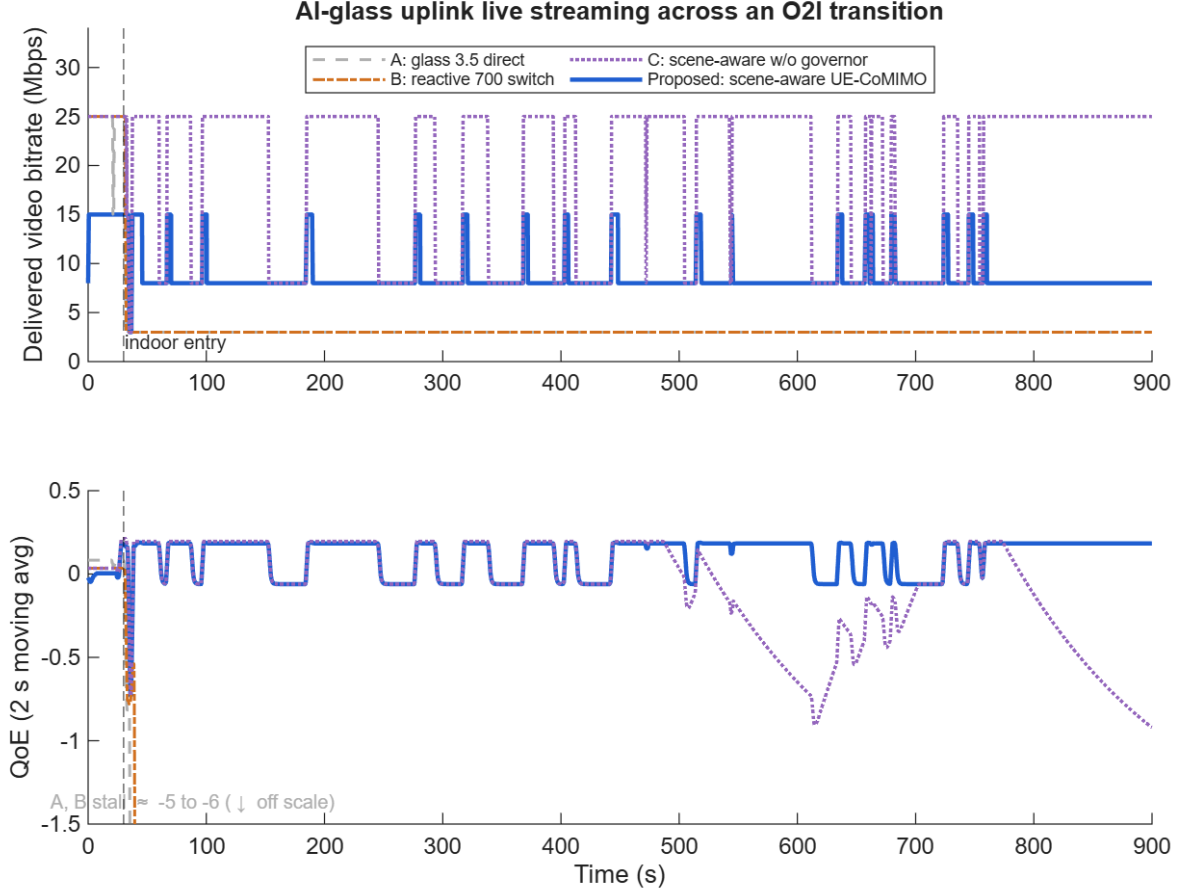


Fig. 1. Scenario 1: delivered video bitrate (top) and smoothed QoE (bottom) across the O2I transition (entrance at $t = 30$ s) and the long indoor dwell. The proposed policy pre-switches before entry, engages the shared CPE, and then encodes content-adaptively: it streams at the 15 Mbps target during motion and the busy CPE spells but drops to 8 Mbps on the static dwell (perceptually equivalent, Section II-F2), so its delivered-bitrate curve alternates between the two levels rather than sitting at the top rung, and its QoE (lower panel) stays at the top throughout. The ablation C keeps chasing the top rung on the CPE; as its glasses heat past the safe limit, its QoE drifts downward over the dwell and falls below the proposed policy, while A and the reactive baseline B both sit at the bottom, stalling in the RF dead-zone that blocks their macro link.

run for $T_{ep} = 2400$ s so that the small glass battery actually depletes and longevity becomes the operative question: a single walk-in then dwell; an indoor $\leftrightarrow$ outdoor shuttle that repeatedly recrosses the O2I boundary; outdoor-only pacing with no CPE; and indoor-only pacing. The three families that enter the venue traverse or dwell in the RF dead-zone of (7), so they also exercise the coverage-hole regime, whereas outdoor pacing never does. Prediction is non-ideal throughout ($\sigma_p = 0.15$). Table IV reports the 12-seed means.

Four observations follow.

(i) *The proposed policy dominates on both metrics everywhere.* P attains both the highest mean QoE and the highest mean quality-seconds in all four trajectory families (and a non-negative mean QoE in three of the four, the exception being a marginal -0.02 in the relentlessly indoor-pacing case). Against its nearest quality-seconds competitor (the ablation C), it delivers 37 to 88% more useful quality-seconds and 37 to 87% more uptime.

(ii) *Raw uptime is a trap; quality-seconds is the right objective.* The reactive baseline B often survives *longest* (up

to 2389 s) yet collapses on quality-seconds, because in the three venue-entering families it stalls in the macro dead-zone it cannot escape: it manages only 33 quality-seconds in the walk-in case and 279 in indoor pacing, against P's 2014 and 1359, while policy A stalls even harder (as few as 1 quality-second). Where there is no dead-zone to fall into (outdoor pacing) B does stream, but still trails P because it cannot reach the target. The lesson is that surviving longer on a dead or low-rate link is worthless; the energy-budget tier (Mechanism 1) instead makes the proposed policy ride the target rung to death on a link that is actually delivering, trading a little raw uptime for substantially more useful quality-seconds.

(iii) *The complexity-aware extensions convert static time into longevity.* In the static-heavy walk-in case, Mechanisms 2 and 3 drop the encoder to 8 Mbps and the camera to ≈ 20 fps during the dwell, extending uptime from 1569 s (governor only) to 2181 s (+39%) at unchanged perceived quality, since the saving attacks the fixed P_{fix} overhead that bitrate alone cannot. Even in the motion-dominated outdoor-pacing case, where little static time exists to exploit, P still wins outright

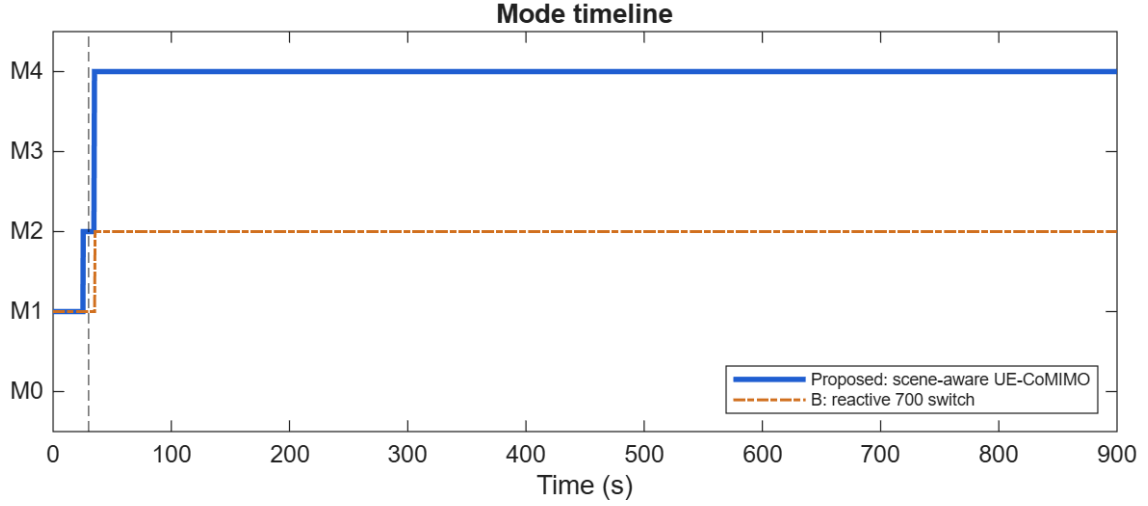


Fig. 2. Scenario 1 mode timeline. The proposed policy pre-switches 3.5 $\rightarrow$ 700 MHz before the entrance, then settles into full delegation (M4, phone cellular off) for the indoor dwell: the dwell lies inside the RF dead-zone, which blocks the phone’s 700 MHz leg and so collapses the packet-split cooperation (M3) onto pure CPE delegation. The reactive baseline switches only after the link has degraded, stays on M2, and then stalls when the dead-zone blocks 700 MHz. (Where the macro 700 MHz survives, e.g. the indoor-pacing family of Table IV, the proposed policy instead alternates M3/M4 with the CPE load.)

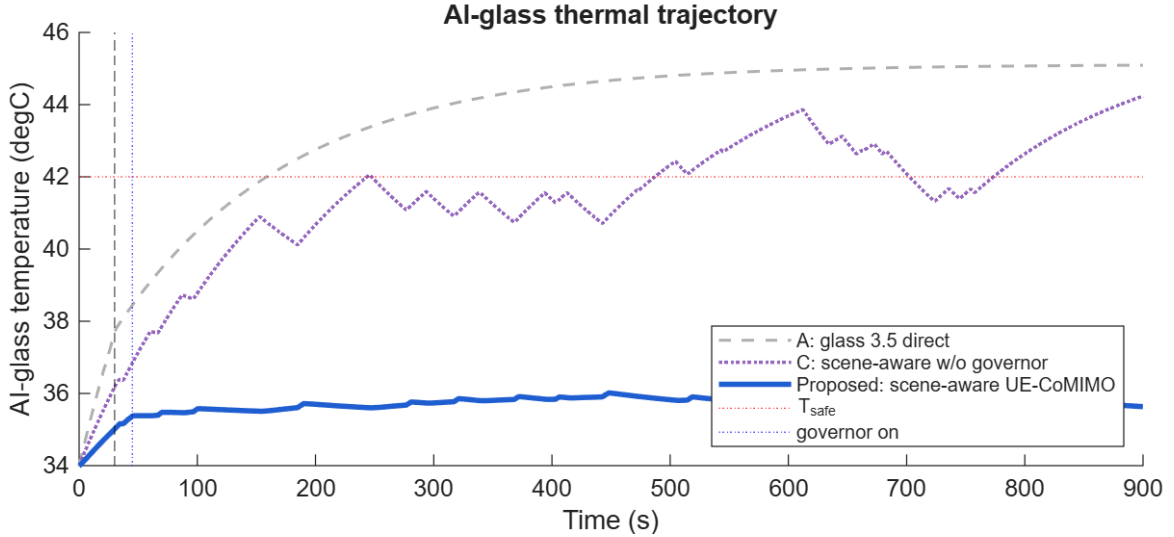


Fig. 3. Scenario 1 AI-glass temperature. Direct cellular (A) climbs toward 45 $^{\circ}\text{C}$, well above the 42 $^{\circ}\text{C}$ safe limit. The ablation C (no governor) crosses the limit partway through the dwell while chasing the top bitrate rung on the CPE and then oscillates above it as the shared CPE load forces intermittent rate drops, peaking at 44.2 $^{\circ}\text{C}$. The proposed policy activates the governor (vertical dotted line) once a long stable indoor dwell is recognized, and the complexity-aware coding then drops the encoder to 8 Mbps and 20 fps on the static dwell, so the temperature settles around 36 $^{\circ}\text{C}$, comfortably below the limit.

because Mechanism 1 avoids the wasteful top-rung streaming that drains the baselines by ≤ 960 s.

(iv) *The advantage is unique to the predictive policy and robust to prediction error.* Glass power depends only on bitrate and frame rate (22), and only the proposed policy adapts both to predicted scene complexity; the baselines waste the top rung at full frame rate on static scenes. The ordering in Table IV is unchanged as the prediction noise is raised to $\sigma_p = 0.3$, confirming that the gain comes from exploiting the static *environment* structure, not from clairvoyance.

K. Limitations

This is a system-level study built on standard link budgets and channel models. It does not re-run a physical-layer MIMO simulation, and the cooperative-rate expressions are tractable abstractions rather than detailed waveform results. The glass and phone power and battery figures are illustrative additive device models, not measured numbers, and the QoE weights encode a design preference rather than a calibrated user study. In particular, the thermal weight makes the skin-safety limit a hard constraint, and the battery-death rule freezes rather than penalizes the QoE average. The scene-complexity model, the linear bitrate- and frame-rate-versus-complexity needs of (19)–(20), and the min-of-adequacies perceptual utility of (21) are

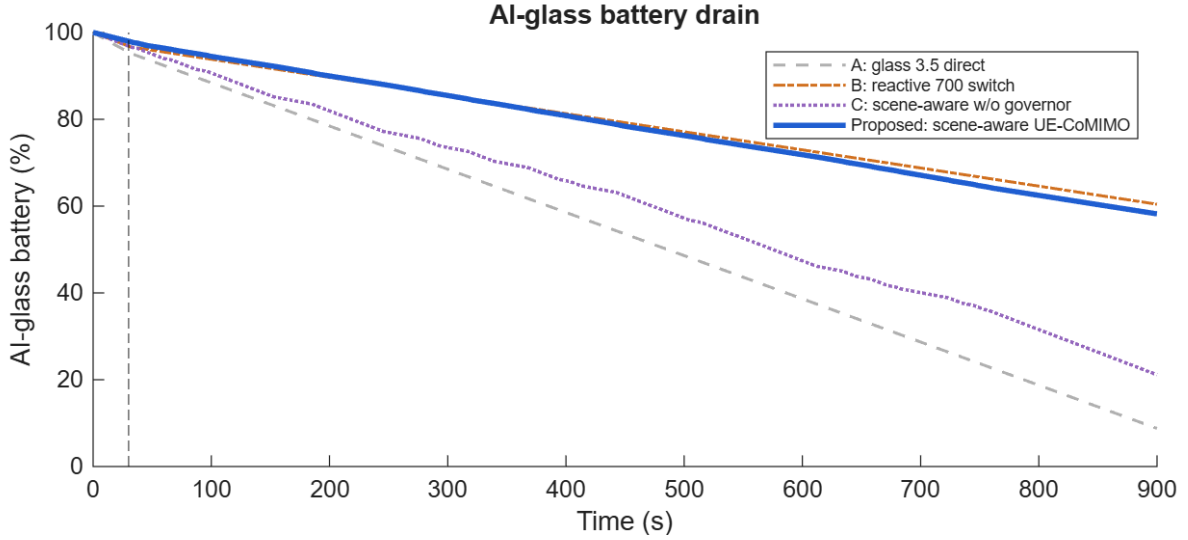


Fig. 4. Scenario 1 AI-glass battery drain. All glasses start full; the slope is set by the streaming power. Direct cellular (A) and the no-governor ablation (C) drain fastest, ending at 8.8% and 21.2%. The proposed policy ends at 58.2%, lasting about 2.8 times as long as C, because it caps the bitrate at the target and then, on the static dwell, encodes content-adaptively at 8 Mbps and 20 fps (Mechanisms 2 and 3), cutting the encoder and camera power at no perceptual cost. The reactive baseline B coasts longest ($\approx 60\%$) only because it is stalling at near-zero rate in the macro dead-zone rather than streaming, illustrating that raw battery life on a dead link is no substitute for sustained useful quality.

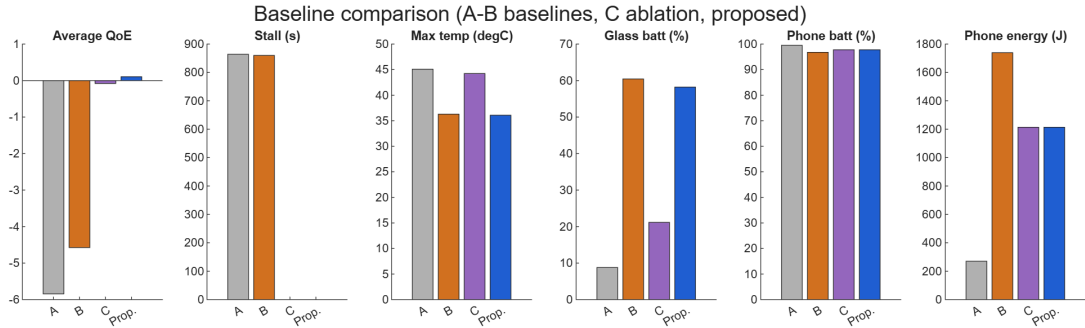


Fig. 5. Scenario 1 aggregate comparison across policies (average QoE, stall time, maximum glass temperature, glass and phone battery remaining, and phone energy). The proposed policy is the only one that is simultaneously stall-free, within the thermal limit, and competitive on both batteries.

deliberately simple stand-ins for a perceptual encoder model and a measured rate-distortion-versus-motion curve. Their role is to expose the qualitative lever (a static scene is cheap to encode and to capture), and the frame-rate power scaling in (22) likewise charges the camera and ISP per captured frame without modeling the full sensor pipeline. The trajectory and complexity prediction is idealized as the true near-future path plus Gaussian error. A deployed system would estimate it from motion sensors and on-device scene understanding, and the robustness to σ_p is the relevant guard. The shared-CPE background load is a two-state Markov abstraction of a much richer multi-user process, and the two Wi-Fi hops of M4 are assumed not to contend for airtime, which holds only when separate channels are provisioned. The make-before-break gap is deliberately conservative relative to standardized NR handover interruption. The battery-longevity findings assume the glasses run on their internal cell. In practice, an external accessory (for example a magnetic power capsule of several Wh) would decouple session length from the internal battery and let the agentic QoE advantage persist indefinitely, and the

interactive implementation includes such an accessory as a toggle. These choices are acceptable for the purpose here, which is to illustrate how an agentic controller converts physical-layer headroom into sustained user-perceived performance, but the absolute numbers should not be read as device specifications.

III. SCENARIO 2: WEARABLE-COLLABORATIVE BLIND-SPOT SENSING AND WARNING

A. Narrative and Motivation

Scenario 2 keeps the same creator on the same electric scooter, but stresses the *sensing* path rather than the uplink. The rider wears AR glasses and a smartwatch, carries a smartphone that acts as the hub agent, and the scooter carries a customer-premises equipment (CPE) node. While the primary service is still first-person streaming, a cyclist approaches from the rear blind spot. The smartphone transmits a probing signal toward the rear; the smartwatch is the default bistatic receiver, and the scooter CPE can be recruited as an auxiliary bistatic receiver. Because the transmitter and receiver are physically separated, this bistatic geometry suppresses self-interference

TABLE IV

MULTI-TRAJECTORY EVALUATION OF THE FULL PROPOSED CONTROLLER AGAINST ALL BASELINES, 12 SEEDS PER CELL, EPISODE $T_{ep} = 2400$ s, NON-IDEAL PREDICTION ($\sigma_p = 0.15$). METRICS ARE THE MEANS OF AVERAGE QoE, GLASS UPTIME (TIME TO BATTERY DEATH, CAPPED AT T_{ep}), AND USEFUL QUALITY-SECONDS (27). THE PROPOSED POLICY (P) ATTAINS BOTH THE HIGHEST MEAN QoE AND THE HIGHEST MEAN QUALITY-SECONDS IN ALL FOUR TRAJECTORY FAMILIES.

Trajectory	Metric	A	B	C	P
Walk-in & dwell	Avg QoE	-5.91	-4.70	-0.15	+0.10
	Uptime (s)	989	2349	1164	2181
	Quality-sec	31	33	1073	2014
Indoor↔outdoor	Avg QoE	-4.37	-0.57	-0.56	+0.03
	Uptime (s)	842	1268	1079	1522
	Quality-sec	366	897	988	1392
Outdoor pacing	Avg QoE	-2.91	-1.28	-1.28	+0.00
	Uptime (s)	806	960	960	1476
	Quality-sec	491	960	960	1475
Indoor pacing	Avg QoE	-6.04	-2.69	-0.29	-0.02
	Uptime (s)	1005	2389	1164	1595
	Quality-sec	1	279	989	1359

relative to a monostatic small-form-factor radar and forms a wider virtual aperture.

The episode contains two successive disturbances on one timeline. First, during part of the approach the rider's arm and body occlude the watch-to-target echo path (a *body blockage*), so the watch goes nearly blind. Second, while the target is still in the rear warning zone, the scooter CPE becomes momentarily unavailable (a link drop, power glitch, or it rolls out of range). These two events are chosen so that *no single fixed cooperation choice covers both*: a watch-only policy fails the blockage, and a policy that statically commits to the CPE fails the failure. The agentic policy instead recruits the CPE for the blockage and, on a micro-agent failure report, re-plans to phone-watch bistatic sensing, maintaining the rear-hazard warning throughout. The control objective couples sensing reliability, uplink streaming, and device energy through a joint QoE that rewards timely warnings and penalizes misses, energy, and mode switching.

B. Geometry and Mobility

The rider body origin is at the scooter. Device positions (meters) are smartphone transmitter $(-0.1, 0.58, 1.2)$, watch receiver $(0.0, 0.58, 1.15)$ in its visible pose and $(0.48, 0.0, 1.15)$ in its body-blocked pose, and scooter CPE $(0.55, -0.62, 0.75)$. The target follows a two-phase trajectory. During the approach $0 \leq t \leq T_a$ with $T_a = 10$ s,

$$\mathbf{p}_{tar}(t) = \mathbf{p}_0 + \frac{t}{T_a}(\mathbf{p}_1 - \mathbf{p}_0), \quad (28)$$

with $\mathbf{p}_0 = (-24, 0, 1.1)$ and $\mathbf{p}_1 = (-8, 0, 1.1)$; afterwards it passes the rider at $v_p = 6$ m/s along $+x$. The episode is 15 s at a 0.1 s step. The rear warning-zone event is

$$\mathcal{Z}_{rear}(t) = \{x_{tar}(t) < 0\} \cap \{\|\mathbf{p}_{tar,xy}(t)\| \leq R_{warn}\}, \quad (29)$$

with $R_{warn} = 18$ m. The target enters the zone at about 3.8 s and leaves it as it passes the rider near 11.3 s.

C. Bistatic Sensing Link Budget

For receiver r the target echo power is

$$P_{echo,r}(t) = \frac{P_{tx} G_r \lambda^2 \sigma_{rcs}}{(4\pi)^3 d_{tx}^2(t) d_{tx,r}^2(t)} 10^{-A_r(t)/10}, \quad (30)$$

where d_{tx} is the phone-to-target distance, $d_{tx,r}$ the target-to-receiver distance, G_r the receiver antenna gain, and $A_r(t)$ the blockage loss on the receiver's echo path.

Why the CPE outperforms the watch. The two bistatic receivers differ in two ways, both of which favor the CPE. First, antenna gain: the wrist-worn watch has $G_w = 2.0$ while the scooter-mounted CPE has $G_c = 4.0$, so by (30) the CPE collects twice the echo power, a 3 dB advantage, at the same geometry. Second, and decisively, the body-blockage loss $A_r(t)$ is nonzero only on the watch echo path: the CPE is mounted on the scooter, away from the rider's torso, and keeps a clear line of sight, whereas the rider's arm and body can occlude the watch. The two effects compound. Even unblocked, the CPE detection-probability curve lies above the watch curve; during the blockage the watch additionally loses up to 20 dB while the CPE is untouched (Fig. 7). This gain-plus-line-of-sight advantage is exactly why recruiting the CPE rescues the sensing task when the watch confidence drops.

Self-interference and direct-path leakage. The probing transmitter and the receivers are never perfectly isolated, and the leakage must be driven below the weak target echo. The monostatic phone receiver, whose transmitter and receiver are co-located, sees a strong self-interference (SI) term suppressed in two stages,

$$P_{SI}^{pre} = P_{tx} 10^{-A_{SI}/10}, \quad (31a)$$

$$P_{SI}^{res} = P_{tx} 10^{-(A_{SI}+D_{SI})/10}, \quad (31b)$$

where A_{SI} is the *pre-ADC analog* suppression (isolation, analog cancellation) and D_{SI} the *post-ADC digital* cancellation. The bistatic watch and CPE receivers instead see a direct-path (DP) leakage from the phone transmitter, set by free-space loss

over the transmitter-to-receiver distance $d_{\text{DP},r}$ and the receiver gain,

$$P_{\text{DP},r} = P_{\text{tx}} G_r \left(\frac{\lambda}{4\pi d_{\text{DP},r}} \right)^2, \quad (32a)$$

$$P_{\text{DP},r}^{\text{res}} = P_{\text{DP},r} 10^{-(A_{\text{DP}} + D_{\text{DP}})/10}, \quad (32b)$$

with A_{DP} and D_{DP} the analog (pre-ADC) and digital (post-ADC) direct-path suppression. The aligned suppression budget is

Interference path	Pre-ADC analog	Post-ADC digital	Total
Monostatic ($A_{\text{SI}}, D_{\text{SI}}$)	65 dB	35 dB	100 dB
Bistatic ($A_{\text{DP}}, D_{\text{DP}}$)	40 dB	60 dB	100 dB

Both paths reach the same 100 dB total, but the *split* is what matters, because only the pre-ADC (analog) stage protects the converter. A b -bit ADC with back-off B_{ADC} has quantization-noise floor

$$P_q = P_{\text{pre}} 10^{-(6.02b + 1.76 - B_{\text{ADC}})/10}, \quad (33)$$

with $b = 10$ and $B_{\text{ADC}} = 6$ dB, so a large residual still present at the ADC input raises the quantization floor by an amount that no post-ADC digital cancellation can later remove. The co-located monostatic SI is by far the strongest interferer, so it is given the most analog headroom (65 dB before the ADC) to keep the converter out of compression; the spatially separated bistatic direct path is already weaker, so 40 dB of analog suppression suffices before the ADC and more of the budget is left to digital cancellation (60 dB).

Collecting the target echo against this disturbance, the mean effective sensing SINR at receiver r is

$$\gamma_r = \frac{\bar{P}_{\text{echo},r}}{P_{\text{th}} + P_{\text{cl}} + P_{\text{res},r} + P_{q,r}}, \quad (34)$$

where $P_{\text{th}} + P_{\text{cl}}$ is the thermal-plus-clutter floor, $P_{\text{res},r}$ is the residual interference (monostatic SI for the phone, residual direct path for the watch and CPE), and $P_{q,r}$ the ADC quantization-noise power of (33). This SINR, not the raw range, is what drives detection: the sensing model maps γ_r through a receiver-specific waveform-CFAR curve (Section III-E). The remaining sensing parameters are listed in Table V.

D. Body Blockage

Body blockage affects only the watch echo, not the CPE link, the monostatic self-interference, or the bistatic direct path. The blockage fraction $\beta(t) \in [0, 1]$ is a smooth ramp-in, dwell, and ramp-out built from the smoothstep $s(u) = u^2(3 - 2u)$: zero before 2.7 s, ramping over $[2.7, 3.1]$, unity over $[3.1, 5.9]$, ramping out over $[5.9, 6.3]$, and zero thereafter. The full blockage loss is $A_{\text{watch}}(t) = 20\beta(t)$ dB, applied to the watch echo power, so it enters the watch SINR of (34) directly,

$$\gamma_w(t) = \frac{\bar{P}_{\text{echo},w}^0(t) 10^{-20\beta(t)/10}}{P_{\text{th}} + P_{\text{cl}} + P_{\text{res},w} + P_{q,w}}, \quad (35)$$

and the watch detection probability is read from its single SINR curve, $P_{d,w}(t) = \mathcal{L}_w(\gamma_{w,\text{dB}}(t))$. Folding the blockage

into the SINR, rather than blending a separate blocked detection curve, avoids counting the loss twice.

E. Waveform-CFAR Detection-Probability Lookup

The detection probabilities are estimated offline by a waveform-level CA-CFAR Monte Carlo and stored as functions of the mean effective SINR of (34), not of range, since SINR is the quantity the link budget actually delivers. For each receiver and each SINR point γ on the sweep $-35:20$ dB, QPSK-OFDM symbols are generated with the target delay at an 18 m reference range and the target amplitude scaled to realize γ ; the waveform then passes through clutter, the receiver-specific residual interference (monostatic SI or bistatic direct path), ADC clipping and quantization, digital cancellation, known-symbol channel estimation, a range IFFT, and a range-1D CA-CFAR, with a shared Swerling-I exponential RCS factor drawn per trial. The empirical detection probability $\hat{P}_{d,r}(\gamma) = \frac{1}{N_1} \sum_i \mathbf{1}\{Z_{r,i} > \eta_{r,i}\}$ uses $N_1 = 2000$ target-present trials, and the CFAR multiplier η is calibrated to a 10^{-4} false alarm over $N_0 = 5000$ target-absent trials; the controller uses the monotonic cumulative-maximum of each curve to suppress finite-sample dips.

Crucially, the curves are *receiver-specific*: equal analytical SINR does not imply equal CFAR statistics, because the phone carries structured residual self-interference while the watch and CPE carry a geometry-dependent direct path, giving different pre-ADC power, clipping, quantization, and calibrated thresholds. As Fig. 7 shows, the CPE reaches $P_d = 0.8$ near -13 dB, the phone near $+1$ dB, and the watch only near $+5$ dB, so at equal SINR the CPE is the most reliable receiver; together with its higher gain and its immunity to body blockage, this is why recruiting it rescues the sensing task. During the episode each receiver token is $P_{d,r}(t) = \mathcal{L}_r(\gamma_{r,\text{dB}}(t) + X_r(t))$, where $\mathcal{L}_r$ is the receiver- r lookup and X_r an AR(1) dB fluctuation; SINR values outside the grid use the nearest endpoint.

F. Agentic Sensing Controller

Let $a_{\text{CPE}}(t) \in \{0, 1\}$ denote whether the CPE is active for sensing. Define the approach-and-degrade event

$$\mathcal{S}(t) = \mathcal{A}(t) \cap \{P_{d,w}(t) < 0.8\}, \quad (36)$$

where $\mathcal{A}(t)$ is the event that the target is behind and approaching. If $\mathcal{S}(t)$ holds for three consecutive steps the CPE is recruited after a 0.4 s setup delay; it is released if $\mathcal{S}(t)$ stays false for 1.0 s. Communication need is logged but does not by itself recruit or retain the CPE.

The new ingredient relative to the standalone model is closed-loop *resilience*. The CPE is available only outside a failure window $[t_{f,0}, t_{f,1}] = [8.0, 10.5]$ s. When the micro-agent reports the failure, the hub cannot recruit the CPE and immediately releases it if active, so $a_{\text{CPE}}(t) = 0$ throughout the window and the agent falls back to watch-only (phone-watch bistatic) sensing. Because the body blockage has cleared by 6.3 s, the watch is healthy again during the failure window, so this fallback restores coverage without a sensing-coverage gap. The agentic intent also switches from streaming quality to

safety-critical sensing while the rear target is present, capping the video enhancement layer and duplicating the warning over the phone path.

G. Detection Fusion and Warning Decision

Before recruitment the agent uses the watch probability directly, $P_{d,\text{agent}}(t) = P_{d,w}(t)$. When the CPE is active, the watch and CPE probabilities are fused by a reliability-limited soft OR,

$$P_{d,\text{agent}}(t) = \min\left[P_{\max}, 1 - (1 - \rho_w P_{d,w})(1 - \rho_c P_{d,c})\right], \quad (37)$$

with $\rho_w = 0.98$, $\rho_c = 0.85$, and $P_{\max} = 0.97$. Warnings are produced from noisy token-level range estimates $\hat{r}(t) = r(t) + \epsilon(t)$ with $\epsilon \sim \mathcal{N}(0, \sigma_r^2)$ and $\sigma_r(t) = 0.45 + 2.0(1 - P_{d,\text{agent}}(t))$; a warning is issued when the target is in the rear zone, $\hat{r}(t) \leq R_{\text{warn}}$, and $P_{d,\text{agent}}(t) \geq 0.6$. Warnings inside the rear zone count as successes; warnings outside it count as false alarms.

H. Communication, Energy, and Joint QoE

The phone and CPE uplinks use a UMa-style path loss $L(d) = 13.54 + 39.08 \log_{10} d + 20 \log_{10} f_c$ and a discounted Shannon rate $R = \eta B \log_2(1 + \Gamma_{\text{lin}})$ with $\eta = 0.55$. The communication mode is C2 (phone only), C3 (phone+CPE when the phone link cannot meet the video target with margin), or C4 (safety-priority multi-link with the video bitrate capped) while a rear target is in the zone and the CPE is active. Energy accounts for phone sensing and communication, watch receive, CPE receive and uplink, and glass capture and encoding, integrated over the episode (Table V).

The per-step joint QoE adds streaming utility and a warning reward and subtracts penalties for misses, false alarms, latency, sensing overhead, energy, and mode switching, with weights $(w_{\text{stream}}, w_{\text{warn}}, w_{\text{miss}}, w_{\text{false}}, w_{\text{lat}}, w_{\text{token}}, w_{\text{energy}}, w_{\text{switch}}) = (1, 2, 3, 0.5, 1, 0.3, 0.5, 0.2)$. The missed-warning weight exceeds the warning reward because the scenario is safety-sensitive. As the headline safety metric we use *warning coverage*, the fraction of the in-zone interval over which the warning probability stays reliable (≥ 0.5). Unlike an “ever-warned” success rate, which saturates at one for every policy once the target is close, coverage exposes the mid-hazard blind windows that the two baselines suffer.

I. Baseline Policies

Three policies share the geometry, link budget, and SINR-to- P_d lookup and differ only in cooperation logic. *No-Coop* senses with the watch alone and never recruits the CPE; it is blind during the body blockage. *Static-CPE* is pre-configured cooperation: it delegates rear sensing to the higher-gain CPE for the whole encounter (whenever the target is behind) and idles the watch receive chain, so its detection probability is the reliability-limited CPE-only value $\rho_c P_{d,c}(t)$. Because it has no closed loop, it is strictly better than watch-only from the start, but it keeps no warm watch fallback: it goes blind during the CPE failure and resumes the watch only after the target has

TABLE V
SCENARIO 2 PARAMETERS. STANDARDS-BASED VALUES FOLLOW COMMON 5G NR FR1 ISAC ASSUMPTIONS; ILLUSTRATIVE VALUES ARE DESIGN CHOICES DOCUMENTED HERE FOR REPRODUCIBILITY.

Parameter	Value
Carrier f_c / bandwidth B	3.65 GHz / 100 MHz
Phone sensing Tx power	23 dBm
Receiver noise figure	7 dB
Target RCS σ_{RCS}	0.1 m ²
Watch / CPE Rx antenna gain G_w/G_c	2.0 / 4.0
Monostatic SI suppr. (pre+post-ADC)	65+35 dB
Bistatic direct-path suppr. (pre+post-ADC)	40+60 dB
ADC resolution / back-off	10 bit / 6 dB
Full body-blockage loss	20 dB
Blockage window (ramp/dwell)	[2.7, 6.3] s
CPE failure window	[8.0, 10.5] s
Warning range R_{warn}	18 m
Recruit trigger / setup / release	$P_{d,w} < 0.8$ / 0.4 / 1.0 s
Fusion $\rho_w, \rho_c, P_{\max}$	0.98, 0.85, 0.97
gNB distance / height	200 / 25 m
Phone / CPE uplink Tx	2.0 W / 23 dBm
Video ladder / target	3, 8, 15, 25 / 15 Mbps
Lookup SINR grid / ref. range	-35:2:20 dB / 18 m
CFAR N_1 / N_0 / P_{fa}	2000 / 5000 / 10^{-4}

passed. *Proposed* is the agentic controller of Section III-F: it recruits the CPE when the watch confidence drops, fuses watch and CPE while both are available, re-plans to phone-watch bistatic on the failure report, and releases the CPE when the watch recovers. Because it keeps the watch and fuses, it is never worse than Static-CPE, and it is the only policy covered through both events.

J. Parameter Settings

K. Results and Discussion

Fig. 6 reports the three policies over the episode, averaged across 1000 Monte Carlo trials; Fig. 8 gives the fuller seven-panel diagnostic. Three observations follow.

(i) *Cooperation rescues the blockage, but only the agent rescues the failure.* During the body blockage the mean watch SINR collapses and its detection probability falls to 0.003, so No-Coop loses the rear-hazard warning. Delegating to (Static-CPE) or recruiting (Proposed, about 0.6 s into the episode) the CPE lifts the detection probability to about 0.835 through the blockage and issues the first warning about 0.10 s after the target enters the zone. When the CPE then fails, Static-CPE, having idled the watch, goes blind in turn, whereas the agent re-plans to phone-watch bistatic and stays covered. Over the in-zone interval the agent holds a reliable warning for 98.7% of the time, against 57.9% for No-Coop (which loses the blockage window) and 65.8% for Static-CPE (which loses the failure window). Static-CPE thus beats No-Coop, because delegating to the CPE from the start covers the far approach and the blockage, but a fixed plan cannot survive the outage.

(ii) *The QoE gap is decisive.* The mean per-step joint QoE is +0.94 for the agent against +0.20 for No-Coop and +0.09 for Static-CPE, a four- to tenfold improvement, while the agent still sustains the 15 Mbps video target (mean selected bitrate 15.8 Mbps) and incurs no stall. The two baselines pay the large missed-warning penalty during their respective blind windows,

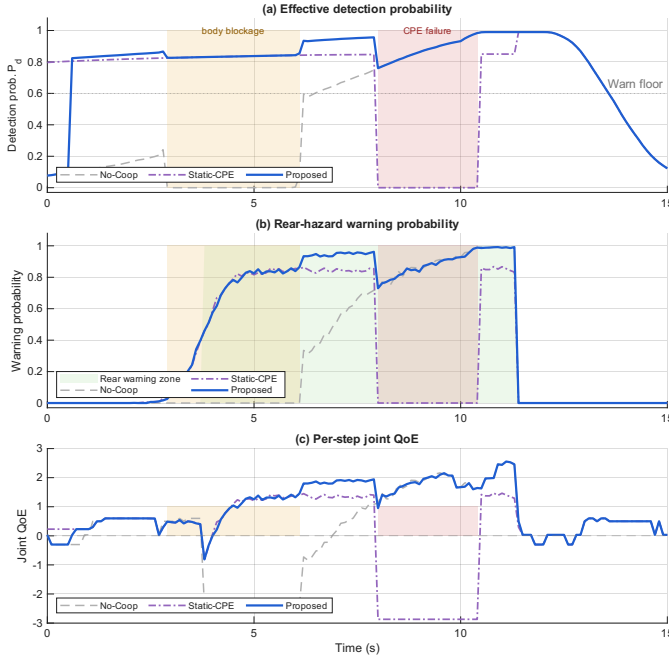


Fig. 6. Scenario 2 policy comparison over the episode (1000-trial mean). The rear target triggers a body-blockage interval (left band) and a CPE-failure interval (right band). (a) effective detection probability, (b) rear-hazard warning probability inside the warning zone, and (c) per-step joint QoE, for No-Coop, Static-CPE, and the proposed agentic policy. No-Coop loses the warning during the blockage and Static-CPE during the CPE failure; only the agent maintains it through both.

visible as the negative excursions in the per-step QoE panel of Fig. 6; Static-CPE additionally pays a standing energy penalty for holding the CPE radio on the whole time, which is why its mean QoE sits below even No-Coop despite better safety coverage.

(iii) *The agent is also more resource-rational than static cooperation.* Because it recruits the CPE only when the watch confidence drops and releases it once the watch recovers, rather than holding it for the whole rear episode, the agent spends less time with the CPE radio engaged than Static-CPE, while achieving strictly better coverage and QoE. The false-alarm rate stays low at 0.022, so the coverage gain does not come from trigger-happy warnings. The headline claim mirrors Scenario 1 in the sensing regime: cooperation restores the capability lost to body blockage, but only intent-aware, closed-loop reconfiguration keeps the safety function alive when a recruited device disappears.

REPRODUCIBILITY

All implementations referenced below, including the browser demos runnable without installation, are published at <https://chaokaiwen.github.io/agnostic-ue-comimo/>. Two implementations accompany this document, and both now carry the full controller, including the predictive energy-aware extensions of Section II-F2 (the scene-complexity model, the energy-budget tier, and the complexity-aware bitrate and frame-rate adaptation). Each mechanism, the prediction-noise level, and the RF dead-zone (7) sit behind their own configuration flags (`cfg.deadzone`/`CFG.deadzone`),

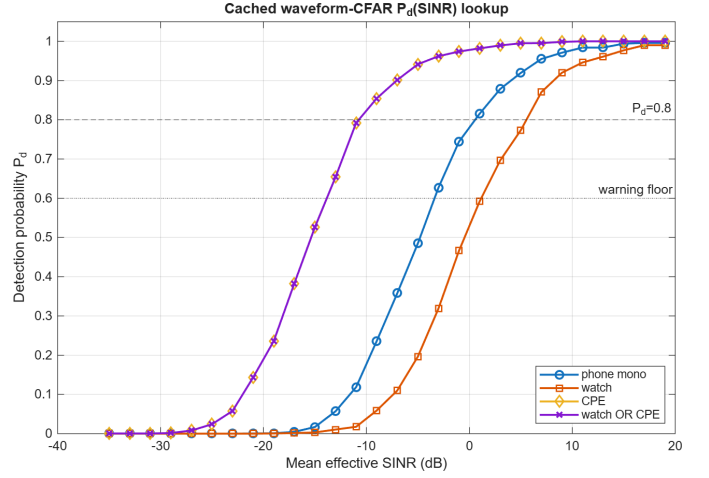


Fig. 7. Offline waveform-CFAR lookup used by the Scenario 2 episode: detection probability $P_d(\gamma)$ versus mean effective SINR γ , estimated separately for the monostatic phone, the watch, and the CPE (the watch-or-CPE fusion overlaps the CPE curve). At equal SINR the CPE reaches $P_d = 0.8$ near -13 dB, the phone near $+1$ dB, and the watch only near $+5$ dB, because equal analytical SINR does not imply equal CFAR statistics.

so the ablations, the noise sweep, and the coverage-hole regime are reproduced by toggling them; with all extensions disabled the per-step utility reduces exactly to the saturating bitrate ratio at the target and the controller reverts to the base scene-aware governor. A single-file MATLAB script, `sim/scene_aware_ue_comimo_uplink/matlab/scene_aware` (base MATLAB only, all parameters in `make_config()`), is the batch reference at seed 7 for the scripted walk-in trajectory ($x_0 = -30$ m to the dwell at x_{stop} over a 900 s episode); with the extensions enabled (the default) it reproduces Table III and Figs. 1 to 5, and disabling them recovers the base scene-aware governor as an ablation. A standalone browser implementation, `game/ue_comimo_game_realistic.html`, ports the identical logic and lets the trajectory be drawn interactively with the mouse, with a $\times 1/\times 10/\times 60$ clock, a live metrics panel and scene-complexity readout, the external power capsule toggle, and an end-of-run export of the per-second time series (HTML report, `.xlsx`, and CSV); Table IV is its 12-seed batch over the four scripted trajectory families. The two implementations use different random-number generators, so individual shadowing realizations differ, but the ordering of policies and every claim above is reproduced.

Scenario 2 is reproduced by the single-file MATLAB script `sim/blind-spot_sensing_warning_ue-comimo_sinr-pd/r` (base MATLAB only, all parameters in `defaultConfig()`). The SINR-to- P_d waveform-CFAR lookup is cached under `cfar_cache/` keyed by a hash of its setup, so a repeated setup loads the table instead of re-running the Monte Carlo; the CPE-failure window sits behind `cfg.cpeFailStartS/cfg.cpeFailEndS` (set the start to `Inf` to disable the failure event). The figures and the per-policy numbers of Section III-K are produced by

```
run_scenario2_sinr_pd_lookup('numTrials', 1000,
    'cfarNumMonteCarlo', 2000, 'cfarNumH0Trials', 5000,
    'outputTag', 'scenario2_matlab_ranged_cf');
```

which writes `blindspot_paper_combined.pdf`

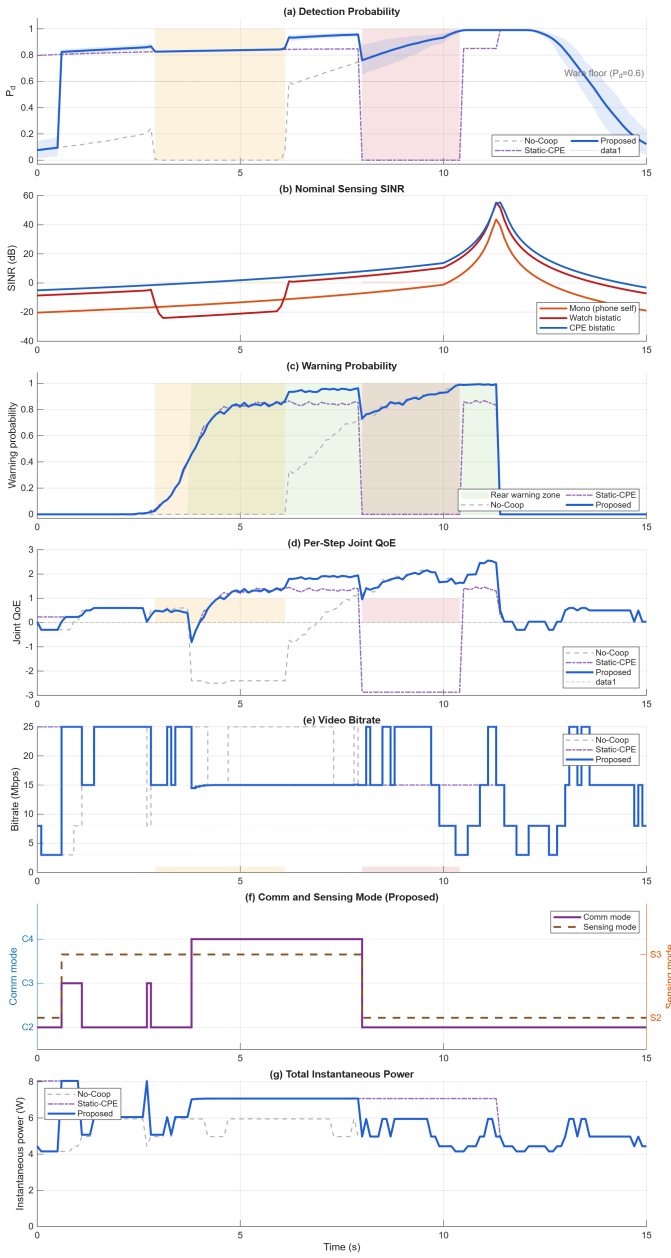


Fig. 8. Scenario 2 seven-panel diagnostic: detection probability, nominal sensing SINR, warning probability, per-step joint QoE, video bitrate, communication/sensing mode (proposed), and total instantaneous power, for the three policies. Shaded bands mark the body blockage (left) and the CPE failure (right).

(Fig. 6), the SINR-to- P_d curves (Fig. 7), the seven-panel overview (Fig. 8), and a JSON metrics summary into its artifact directory. An interactive browser version, sim/blind-spot_sensing_warning_ue-comimo_sinr-pd/game/ue_comimo_blindspot_game_sinr.html, ports the same SINR link budget, lookup curves, and three policies to a top-down radar with a draggable timeline and manual blockage/CPE-failure triggers.